

Amit Kumar Mohanty

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 1+ year of experience building and deploying production-grade AI systems, model-serving platforms, and event-driven applications. Proven track record of leading technical initiatives, designing low-latency services at scale, and delivering measurable business impact through ML engineering and AI-driven automation.

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurugram, Haryana

May 2025 – Present

- Awarded “Rookie of the Year 2025” for surpassing all performance KPIs through advanced model training and data-pipeline optimization.
- Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability.
- Refined an agentic browser-automation AI that autonomously navigates websites to complete real-world tasks, delivering client-ready model iterations.
- Led quality-control and evaluation of the agentic system with a global team, running structured test reviews and daily syncs to validate task accuracy before delivery.
- Designed and orchestrated multi-agent workflows with LangChain and LangGraph, enabling autonomous planning, tool use, and self-correction across chained tasks.
- Built custom data pipelines and curated instruction datasets, delivering 85% of self-generated training data to production as the first member of a global team.

Artificial Intelligence Intern, Infosys – Remote

Nov 2024 – Feb 2025

- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and elevating customer-AI interaction quality.
- Applied hands-on LangChain and LangGraph expertise to orchestrate and deploy production LLM workflows, strengthening end-to-end deployment quality.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing hallucinations by 20%.

PROJECTS

Dynamic Screen Companion

- Built a real-time multi-modal assistant integrating the Gemini API and a high-performance FastAPI backend; optimized OCR pipelines to 95%+ accuracy, reducing manual task time by 40%.
- Implemented concurrent inference streaming with live voice feedback for smooth, hands-free assistance at low latency.

Diabetic Optiscan

- Led a Vision Transformer (ViT-B/16) model on the APTOS dataset with a custom wavelet technique that boosted detection precision by 50%; co-authored a peer-reviewed paper.
- Applied image preprocessing and augmentation on high-resolution retinal scans to improve model generalization and diagnostic consistency.

Currency Detection App for Visually Impaired

- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud.
- Optimized the model for on-device inference with TensorFlow Lite, enabling fast, offline real-time recognition on low-end smartphones.

SKILLS

Languages & Frameworks: Python, C++, SQL, R, FastAPI, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph, LangSmith

AI & Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Deep Learning, Transformers, Hugging Face, BERT, NLP, Prompt Engineering, Model Evaluation & Optimization

MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Model Serving, Model Monitoring, Data Pipelines, Git, Vector Databases (FAISS/Chroma)

Data & Cloud Tools: AWS, GCP (Vertex AI), Power BI, MySQL, MongoDB, Pandas, NumPy, Statistical Analysis

Spoken Languages: English, Hindi, German (A1)

EDUCATION

BITS Pilani

Jul 2026 – Jul 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

K.R. Mangalam University

Jul 2021 – Jul 2025

B.Tech in Computer Science & Engineering (Specialization in AI & ML) | GPA: 8.5/10

CERTIFICATIONS

Data Science Professional – Ducat India

Apr 2024 – Apr 2025

- In-person, project-based training in machine learning, data analysis, and deep learning.

AI/ML for Geodata Analysis – ISRO

- Applied AI/ML to satellite imagery and geospatial data for terrain classification and remote-sensing analysis.